

2 Lecture 2

4. Matrix transposition

Definition 2.1. The transpose of m -by- n matrix $A = [a_{ij}]$ is n -by- m matrix, denoted A^T , whose (i, j) -th entry, $[A^T](i, j)$, is

$$\begin{aligned} [A^T](i, j) &\stackrel{\text{def}}{=} [A](j, i) \\ &= a_{ji}. \end{aligned} \tag{2.1}$$

That is, the transpose A^T of a matrix A can be obtained by any one of the following equivalent actions:

- write the rows of A as the columns of A^T ;
- write the columns of A as the rows of A^T ;
- reflect A over its main diagonal.

For example:

$$\begin{bmatrix} 1 & 3 & 7 \\ 0 & 2 & 5 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 7 & 5 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}^T = [a_1 \quad a_2 \quad \dots \quad a_n].$$

Statement 2.1. *The following rules are satisfied by the operations of matrix addition, scalar multiplication, matrix multiplication and matrix transposition:*

$$1) \quad (A^T)^T = A \quad \text{for any } A \in \text{Mat}(m, n, \mathbb{F}); \tag{2.2}$$

$$2) \quad (\lambda A)^T = \lambda A^T \quad \text{for any } A \in \text{Mat}(m, n, \mathbb{F}) \text{ and } \lambda \in \mathbb{F}; \tag{2.3}$$

$$3) \quad (A + B)^T = A^T + B^T \quad \text{for any } A, B \in \text{Mat}(m, n, \mathbb{F}); \tag{2.4}$$

$$4) \quad (AB)^T = B^T A^T \quad \text{for any } A \in \text{Mat}(m, n, \mathbb{F}) \text{ and } B \in \text{Mat}(n, p, \mathbb{F}). \tag{2.5}$$

Proof. 1), 2) and 3) The proof of these statements is left as an exercise.

4) By Definition 1.5, it suffices to prove that the corresponding entries of the matrices $(AB)^T$ and $B^T A^T$ are the same. Indeed, for any $i \in [p]$ and $j \in [m]$ we have:

$$\begin{aligned} [(AB)^T](i, j) &\stackrel{(2.1)}{=} [(AB)](j, i) \\ &\stackrel{(1.4)}{=} \sum_{k \in [n]} ([A](j, k) \cdot [B](k, i)) \\ &= \sum_{k \in [n]} ([B](k, i) \cdot [A](j, k)) \\ &\stackrel{(2.1)}{=} \sum_{k \in [n]} ([B^T](i, k) \cdot [A^T](k, j)) \\ &\stackrel{(1.4)}{=} [B^T A^T](i, j). \end{aligned}$$

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5. Inverse of a Matrix

Definition 2.2. Suppose that $A \in \text{Mat}_n(\mathbb{F})$. Then

- 1) a matrix $B \in \text{Mat}_n(\mathbb{F})$ is said to be a left inverse of A if $BA = I_n$;
- 2) a matrix $B \in \text{Mat}_n(\mathbb{F})$ is said to be a right inverse of A if $AB = I_n$;
- 3) a matrix A is said to be invertible (or non-singular, or is said to have an inverse) if there exists a matrix $B \in \text{Mat}_n(\mathbb{F})$ such that $AB = BA = I_n$. Such B is called an inverse of A .

For example, one may check that if

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{6} & -\frac{1}{6} & \frac{2}{3} \\ -1 & 0 & 1 \end{bmatrix}$$

then $AB = BA = I_3$.

Lemma 2.1. Let $A \in \text{Mat}_n(\mathbb{F})$. Suppose that there exist $B, C \in \text{Mat}_n(\mathbb{F})$ such that $AB = I_n$ and $CA = I_n$ then $B = C$.

Proof. Note that $B = I_n B = (CA)B = C(AB) = CI_n = C$. ■

Lemma 2.1 implies

Corollary 2.1. Whenever $A \in \text{Mat}_n(\mathbb{F})$ is invertible, its inverse is unique.

The inverse of an invertible matrix A is denoted by A^{-1} , that is, $AA^{-1} = A^{-1}A = I_n$.

Statement 2.2. Let $A, B \in \text{Mat}_n(\mathbb{F})$ be invertible. Then

- 1) $(A^{-1})^{-1} = A$;
- 2) $(AB)^{-1} = B^{-1}A^{-1}$;
- 3) $(A^T)^{-1} = (A^{-1})^T$.

Note that even if A, B and $A + B$ are invertible, there is no “good” formula for $(A + B)^{-1}$.

Proof. 1) Let $B = A^{-1}$ be the inverse of A . Then $AB = BA = I_n$. Thus, by definition, B is invertible and $A = B^{-1}$. Therefore, $A = (A^{-1})^{-1}$.

2) Since

$$AB(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n$$

and

$$(B^{-1}A^{-1})AB = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n,$$

by Corollary 2.1, we have $(AB)^{-1} = B^{-1}A^{-1}$.

3) Since

$$A^T(A^{-1})^T \stackrel{(2.5)}{=} (A^{-1}A)^T = I_n^T = I_n \quad \text{and} \quad (A^{-1})^T A^T \stackrel{(2.5)}{=} (AA^{-1})^T = I_n^T = I_n,$$

by Corollary 2.1, we have $(A^T)^{-1} = (A^{-1})^T$. ■

Item 2 of Statement 2.2 implies

Corollary 2.2. Whenever $A_1, \dots, A_n \in \text{Mat}_n(\mathbb{F})$ are invertible, the matrix $A_1 A_2 \cdots A_n$ is also invertible and

$$(A_1 A_2 \cdots A_n)^{-1} = A_n^{-1} \cdots A_2^{-1} A_1^{-1}.$$

We will return to the study of inverse matrices later (see Theorems ?? and ??).

6. The Trace of a Matrix

Definition 2.3. The trace of an n -by- n matrix $A = [a_{i,j}]$, denoted $\text{tr}(A)$ (or $\text{tr } A$), is defined to be the sum of the entries on the main diagonal of A , i.e.,

$$\begin{aligned}\text{tr } A &\stackrel{\text{def}}{=} \sum_{i \in [n]} [A](i, i) \\ &= \sum_{i \in [n]} a_{ii} = a_{11} + \cdots + a_{nn}.\end{aligned}\tag{2.6}$$

For example:

$$\text{tr} \begin{bmatrix} 1 & 0 & 5 \\ 3 & -2 & -1 \\ 7 & 5 & 4 \end{bmatrix} = 1 + (-2) + 4 = 3.$$

Statement 2.3. *Properties of the trace of a matrix:*

- 1) $\text{tr}(A + B) = \text{tr } A + \text{tr } B$ for any $A, B \in \text{Mat}_n(\mathbb{F})$;
- 2) $\text{tr}(\lambda A) = \lambda \cdot \text{tr } A$ for any $A \in \text{Mat}_n(\mathbb{F})$ and $\lambda \in \mathbb{F}$;
- 3) $\text{tr}(A^T) = \text{tr } A$ for any $A \in \text{Mat}_n(\mathbb{F})$;
- 4) $\text{tr}(AB) = \text{tr}(BA)$ for any $A \in \text{Mat}(m, n, \mathbb{F})$ and $B \in \text{Mat}(n, m, \mathbb{F})$.

Proof. 1), 2) and 3) The proof of these statements is left as an exercise.

4) We have:

$$\begin{aligned}\text{tr}(AB) &\stackrel{(2.6)}{=} \sum_{i \in [m]} [AB](i, i) \\ &\stackrel{(1.4)}{=} \sum_{i \in [m]} \left(\sum_{j \in [n]} [A](i, j) \cdot [B](j, i) \right) \\ &= \sum_{i \in [m]} \left(\sum_{j \in [n]} [B](j, i) \cdot [A](i, j) \right) \\ &= \sum_{j \in [n]} \left(\sum_{i \in [m]} [B](j, i) \cdot [A](i, j) \right) \\ &\stackrel{(1.4)}{=} \sum_{j \in [n]} [BA](j, j) \\ &\stackrel{(2.6)}{=} \text{tr}(BA).\end{aligned}$$

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